Internship at Visteon

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Manager - Ashokkumar Balasubramanian

June 2026

VHAL Team

June 2026

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Introduction

What is VHAL?

Vehicle Hardware Abstraction Layer (VHAL) acts as the vital software bridge between a car's core mechanical hardware (sensors, ECUs) and its digital operating system—most commonly Android Automotive. It translates low-level vehicle signals into standardized digital data so apps can safely read car states and control physical features.

Key Roles and Functions

- Hardware Abstraction
- Reading Vehicle Data
- Executing Commands
- Power Management

How VHAL Works

The VHAL functions as a property system where applications communicate by requesting or altering specific, predefined parameters.

- **Read:** An app requests the current value of a vehicle property (e.g., INFO_EV_BATTERY_CAPACITY for electric cars).
- **Write:** An app sets a value to trigger a physical change (e.g., setting FUEL_DOOR_OPEN to true).
- **Subscribe:** Apps (like the dashboard speedometer) register a callback so the VHAL automatically notifies them the moment a value changes (e.g., CURRENT_GEAR or speed).

Roles and Responsibilities

Roles and Responsibilities

January and February

- Completed following online courses assigned in Cornerstone:
  - Company and Product Overview
  - Behavioral Curriculum

- Participated in interactive training sessions on:
  - Automotive Product Overview
  - Design Patterns - UML
  - Automotive Networks
  - Embedded C for Automotive Industry
  - Automotive Coding Standards
  - System Validation Training

Roles and Responsibilities

March and April

- Participated in interactive training sessions on:
  - GIT/GIT Lab, Artifactory & Conan, DevOps Flow, BuildNext
  - Static Code Analysis - Klockwork & Unit Testing – VectorCast
  - JIRA ,Xray, R4J, EasyBI
  - VBOS & VPDS ,Vistwary
  - C Language Training – 15 sessions

- Studied advanced C++ topics – suggested by Vikash sir
  - Core C++ (Pointers and References, Smart Pointers, RAII concept)
  - OOPs (Inheritance, Polymorphism, Virtual Functions, Interfaces and Abstract Classes)
  - STL (Vector, Map, Set, Array, Algorithms – find, sort, etc.)
  - Multithreading (std::thread, mutex, lock_guard, unique_lock, conditional_variable, atomic)
  - Makefiles

  Presented on these topics to Tanay sir and Vikash sir

Roles and Responsibilities

March and April

- C Hackathon organized by SKILL LYNC
  - Problem statement was based on real life use case
  - Received the first prize among 17 other teams

Roles and Responsibilities

May and June

- Studied Android Basics – suggested by Vikash sir
    - Basics of Android System Architecture
    - Android Boot and Framework Basics
    - System Services Basics
    - Binder IPC Intro Level
    - Android Build System Basics
    - Android File Structure
    - HAL beginner concepts
    - AOSP Build, Flash and Debug
    - Logging and Debugging Tools
    - Permissions and SELinux

- C++ Language Training – 15 sessions

- C++ Hackathon organized by SKILL LYNC

Key Projects & Contributions

Key Projects & Contributions

- Android Studio Projects:
  - Activity Lifecycle Tracker
  - Service Lifecycle Tracker
  - Broadcast Receiver
  - AIDL & HIDL Calculator
- RoboFit Automation Framework
- AI Automation Project

Skills and Knowledge Gained

Skills and Knowledge Gained

- C and C++ advanced knowledge
- Domain Specific knowledge
- Hardware exposure
- Team work and problem solving
- Command line tools(adb, logcat, memsys, meminfo, sysdump)
- GitHub and GitLab access

Visteon®

AI-Based Work
& Implementation

AI-Based Work and Implementation

AI Automation Project

Goal: To build a automated pipeline where logs are processed and parsing of data takes place

Purpose: Easier for the developer to understand logs, accurate value representation and relation to other files

Implementation
- AI models have been downloaded and trained on pre-processed data
- Various models have been tried and tested(1b, 3b, 8b Models)
- 3b models have shown good balance between accuracy and hardware requirements
- Current 3b model gives ~85% accuracy on log parsing
- Relates to other data(message catalogue, canDBC) for better understanding

Problems Faced: Data loss or wrong data extraction on logs with long payload data

Reflections and Learnings

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Throughout our internship, we have gained the following insights:

- Android Overview
- VHAL introduction
- C and C++ applications
- AI in automation
- Overall Company project areas and technologies used
- Hardware exposure
- Team collaboration

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THANK YOU